Statistical distributions – from dice to atomic nuclei

A single throw of a die is unpredictable, but a thousand throws are not. The same goes for a radioactive decay, the speed of a molecule, a measurement error or a child's height. Probability theory describes how random outcomes are distributed, and a few distributions turn up everywhere in nature and society: the binomial, the Poisson, the exponential and above all the normal distribution. Here they are with examples and interactive figures – together with the tools used to draw conclusions from data.

Chance and distributions

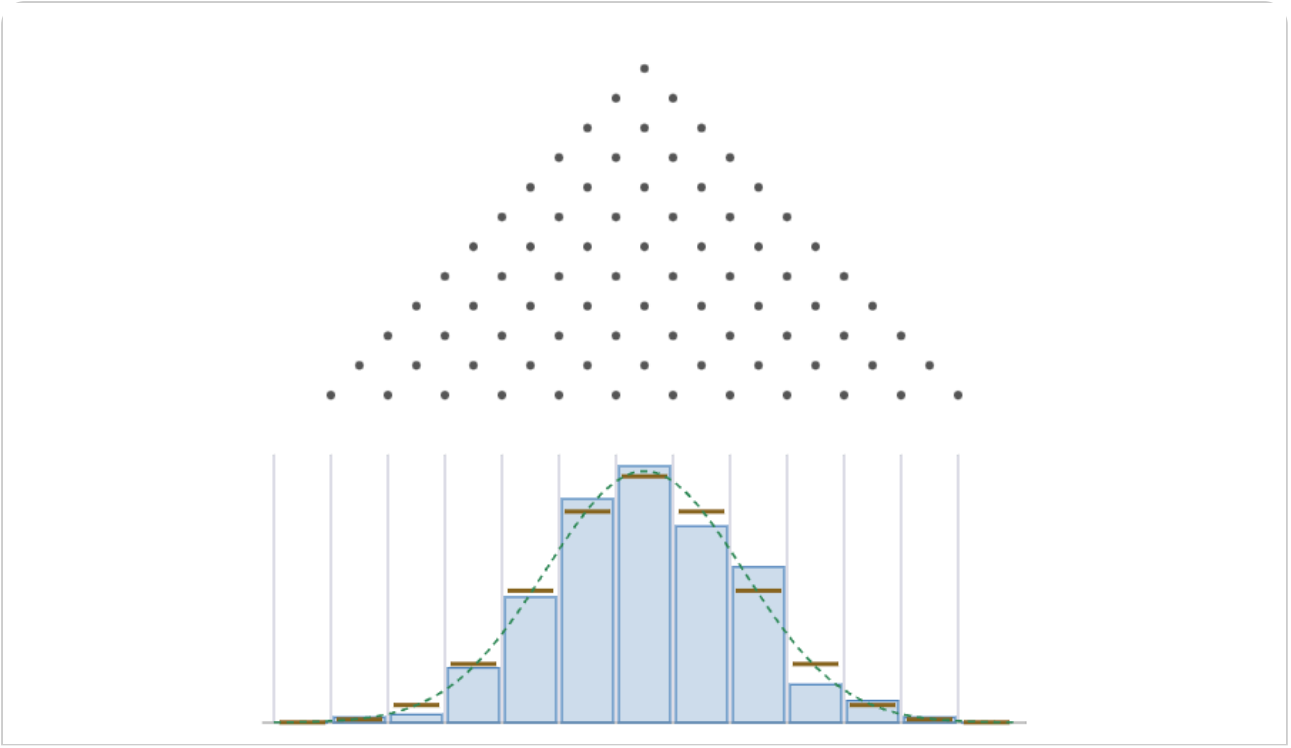
A *random variable* X is a number that depends on a random outcome: the number of pips on a die, the number of decays in a second, a person's height. Its distribution tells us how likely the different values are. For a *discrete* variable we give the probability $P(X = k)$ of each value; for a *continuous* variable a density function $f(x)$, where the probability of an interval is the area under the curve:

$$P(a < X < b) = \int_a^b f(x)\,dx, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

Two numbers describe most of it: the *mean* (expected value) μ, which tells where the distribution lies, and the *standard deviation* σ, which tells how wide it is. The variance is σ^2:

$$\mu = E(X) = \sum_k k\,P(X = k), \qquad \sigma^2 = E\big[(X-\mu)^2\big] = E(X^2) - \mu^2$$

Probability theory began with games of chance: Pascal and Fermat corresponded about dice games in 1654, and Jacob Bernoulli's *Ars Conjectandi* (1713) proved the first version of the law of large numbers. In the 1870s Francis Galton built a board of pins on which balls bounce right or left at each pin – a mechanical machine that shows how order arises from chance.

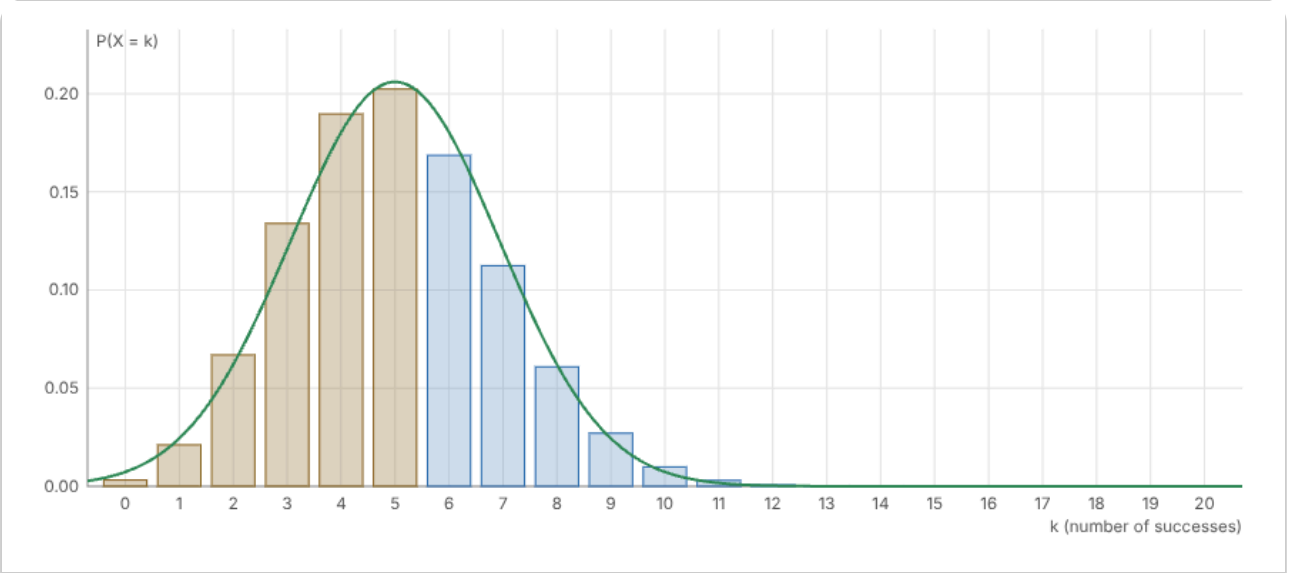


The binomial distribution

Repeat an experiment n times independently, where each trial succeeds with probability p. The number of successes X is *binomially distributed*:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mu = np, \qquad \sigma = \sqrt{np(1-p)}$$

The binomial coefficient $\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$ counts the number of ways the k successes can be placed among the n trials – these are the numbers in Pascal's triangle, and the number of routes to each bin on Galton's board. Examples: the number of tails in 10 coin tosses, the number of correct answers when guessing in a multiple-choice test, the number of boys among 100 newborns (the proportion of boys is about 51.4 %), the number of defective items in a sample.



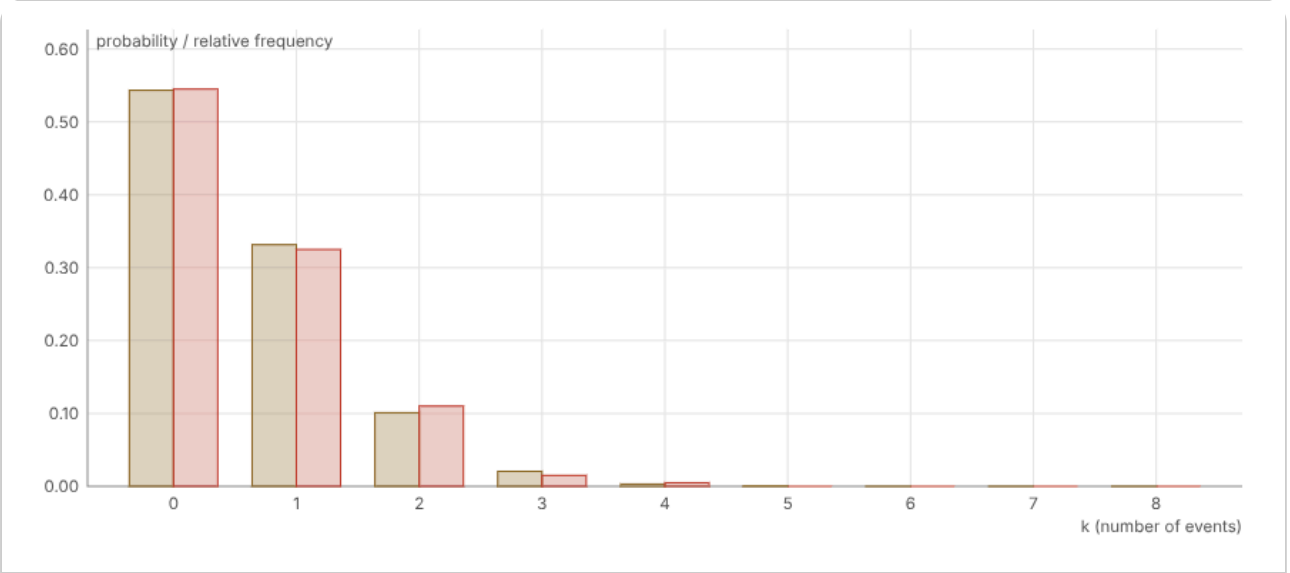
The Poisson distribution – the law of rare events

If n is very large and p very small while $np = \lambda$ is held fixed, the binomial distribution turns into the *Poisson distribution*, derived by Siméon Denis Poisson in 1837:

$$P(X = k) = \frac{\lambda^k}{k!}\, e^{-\lambda}, \qquad \mu = \lambda, \qquad \sigma^2 = \lambda$$

It describes the number of events in a time interval or a region when the events occur independently of each other at a fixed average rate: decays in a radioactive source, photons in a detector, shooting stars per hour, calls to an emergency centre, mutations in a gene. Its hallmark is that the variance equals the mean – which is why the relative uncertainty of a count is $\sqrt{N}/N = 1/\sqrt{N}$. Count 100 decays and the uncertainty is 10 %; count 10,000 and it is 1 %.

Two classic data sets show the law in real life. In 1898 the Russian-German statistician Ladislaus von Bortkiewicz counted the soldiers in the Prussian army killed by horse kicks: 10 army corps over 20 years, 200 corps-years in all. In 1910 Ernest Rutherford and Hans Geiger counted α particles from a polonium source in 2608 time intervals of 7.5 seconds.

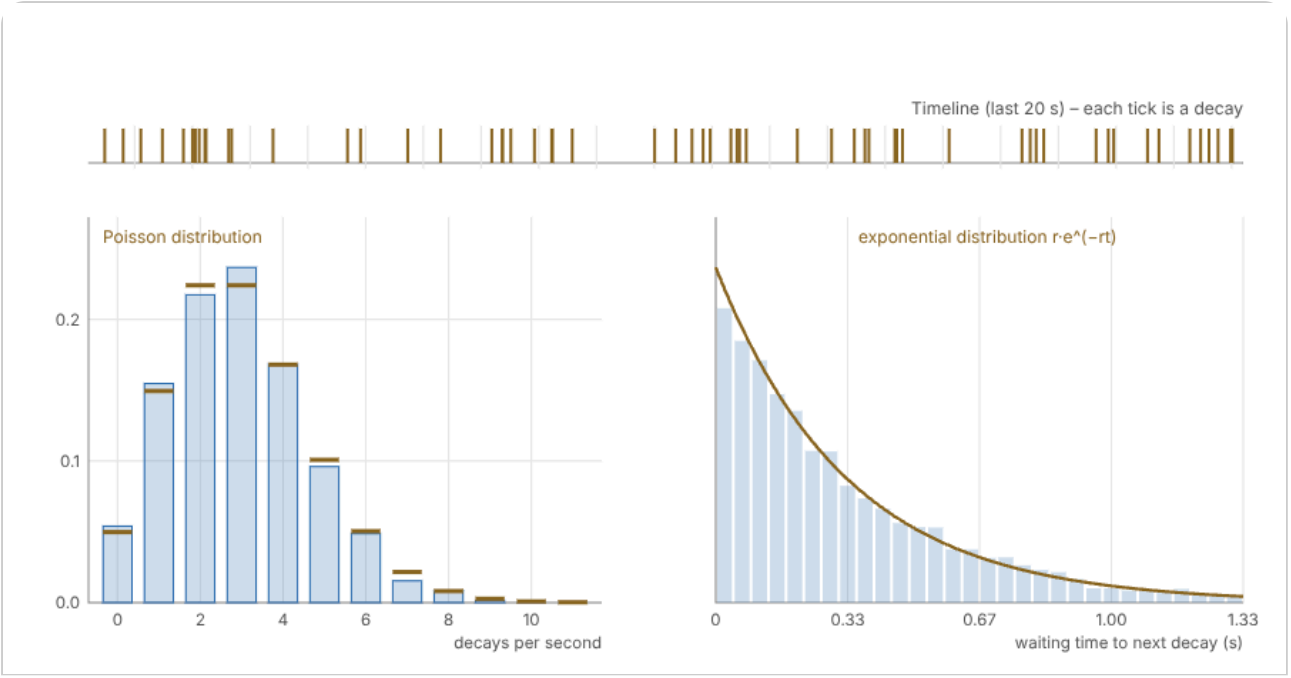


Waiting times and the exponential distribution

Events that occur randomly and independently at rate r form a *Poisson process*. The number in a time t is Poisson-distributed with $\lambda = rt$, and the waiting time T until the next event is *exponentially distributed*:

$$f(t) = r\,e^{-rt}, \qquad P(T > t) = e^{-rt}, \qquad E(T) = \frac{1}{r}$$

The exponential distribution is *memoryless*: if you have already waited 10 seconds, the expected remaining waiting time is still $1/r$. An atomic nucleus does not grow "old" – it has the same probability of decaying in the next second, however long it has existed. That is why a collection of nuclei decays exponentially, $N(t) = N_0 e^{-rt}$, with half-life $T_{1/2} = \ln 2 / r$. The same distribution describes the time between customer arrivals in a queue and the lifetime of electronic components that do not wear out.



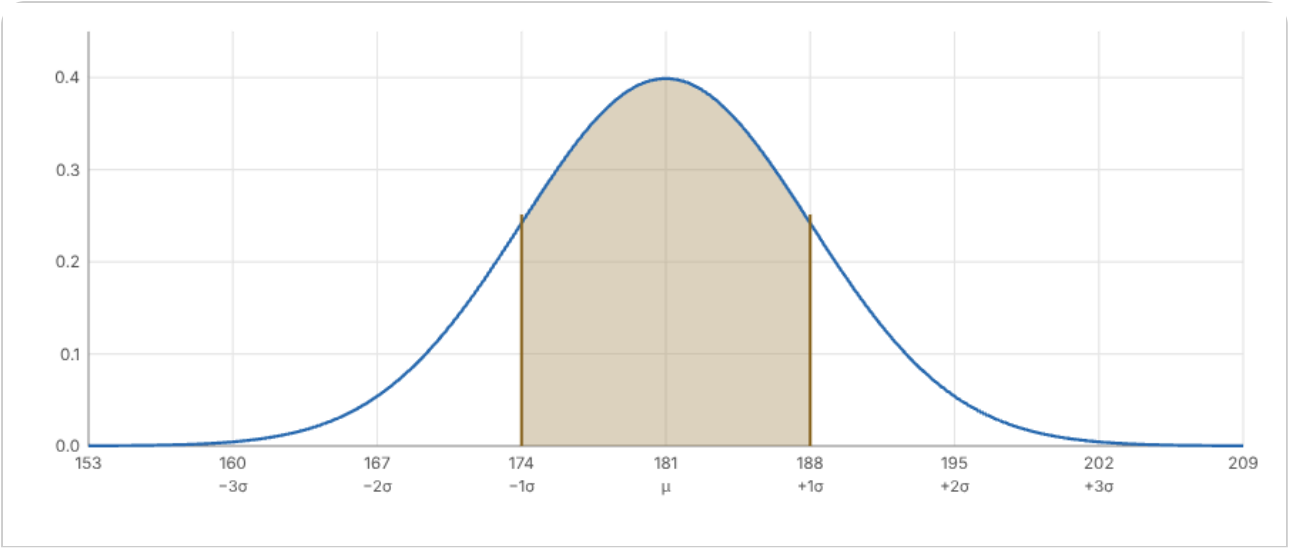
The normal distribution

In 1733 Abraham de Moivre discovered that for large n the binomial distribution approaches a particular bell-shaped curve. Carl Friedrich Gauss used it in 1809 to describe measurement errors in astronomy, and Pierre-Simon Laplace showed in 1810 why it appears so often (see the central limit theorem below). The density is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

All normal distributions are the same curve, merely shifted and stretched. That is why one calculates in *standardised* units $z = (x-\mu)/\sigma$ and remembers the rule of thumb:

Interval	Fraction inside	Fraction outside
μ ± 1σ	68.27 %	about 1 in 3
μ ± 2σ	95.45 %	about 1 in 22
μ ± 3σ	99.73 %	about 1 in 370
μ ± 5σ	99.99994 %	about 1 in 1.7 million

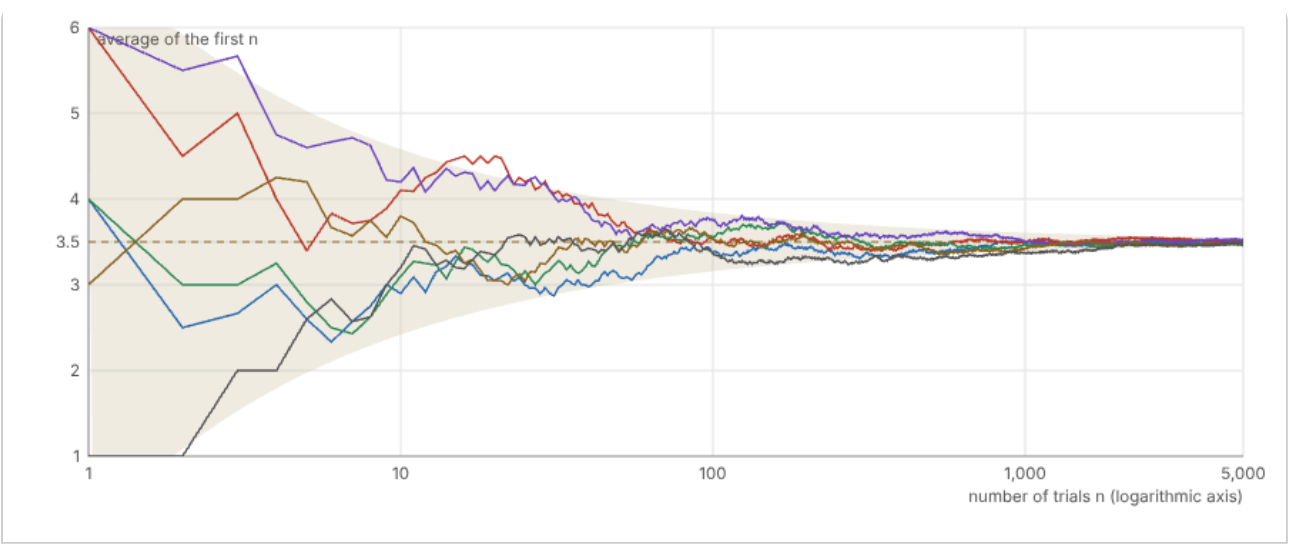


The law of large numbers

The average of many independent outcomes approaches the mean: $\bar{X}_n = \frac{1}{n}(X_1 + \cdots + X_n) \to \mu$. That is why a casino always wins in the long run, why insurance companies can calculate premiums, and why repeated measurements give a better result. But how fast? The standard deviation of the average is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

To halve the uncertainty you must therefore measure four times as often. The law has a condition that is easily overlooked: the distribution must *have* a mean.

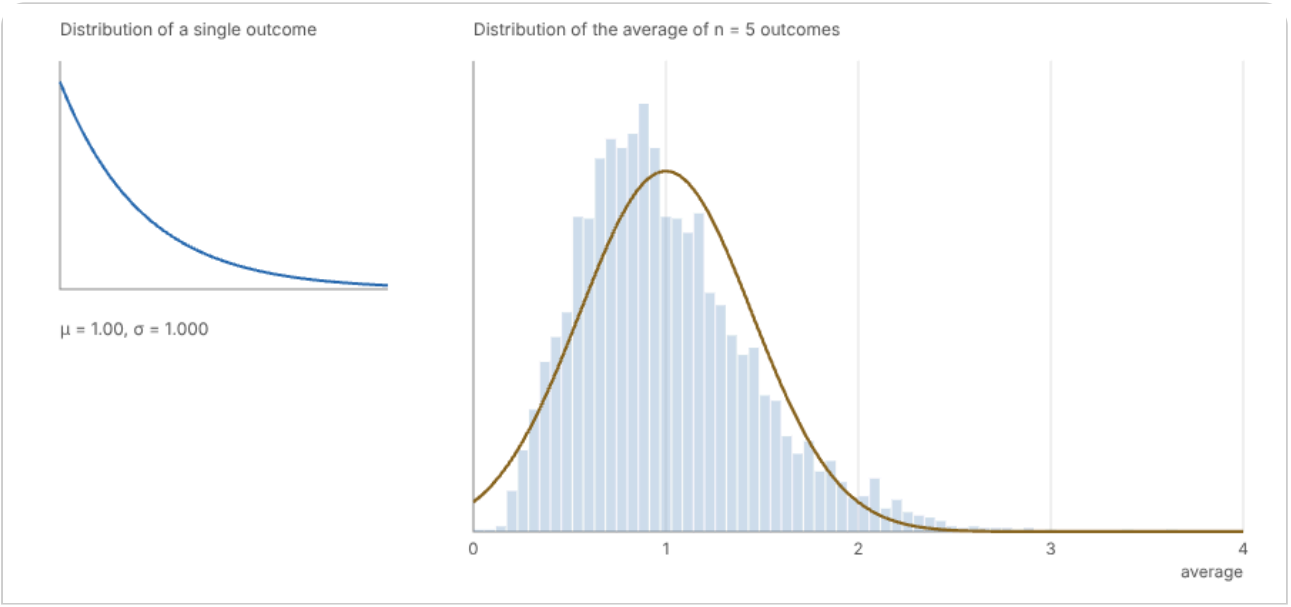


The central limit theorem

The most important theorem in probability theory says that the sum or average of many independent random contributions becomes normally distributed – *almost regardless* of how the individual contribution is distributed, provided it has finite variance:

$$\bar{X}_n \approx N\left(\mu,\ \frac{\sigma^2}{n}\right) \qquad \text{for large } n$$

Laplace proved a version in 1810, and Aleksandr Lyapunov gave a rigorous proof under general conditions in 1901. The theorem explains why the normal distribution is everywhere: a measurement error is the sum of many small disturbances (vibrations, temperature, reading), a person's height results from many genes and environmental factors, and a velocity component of a gas particle is the sum of countless collisions.



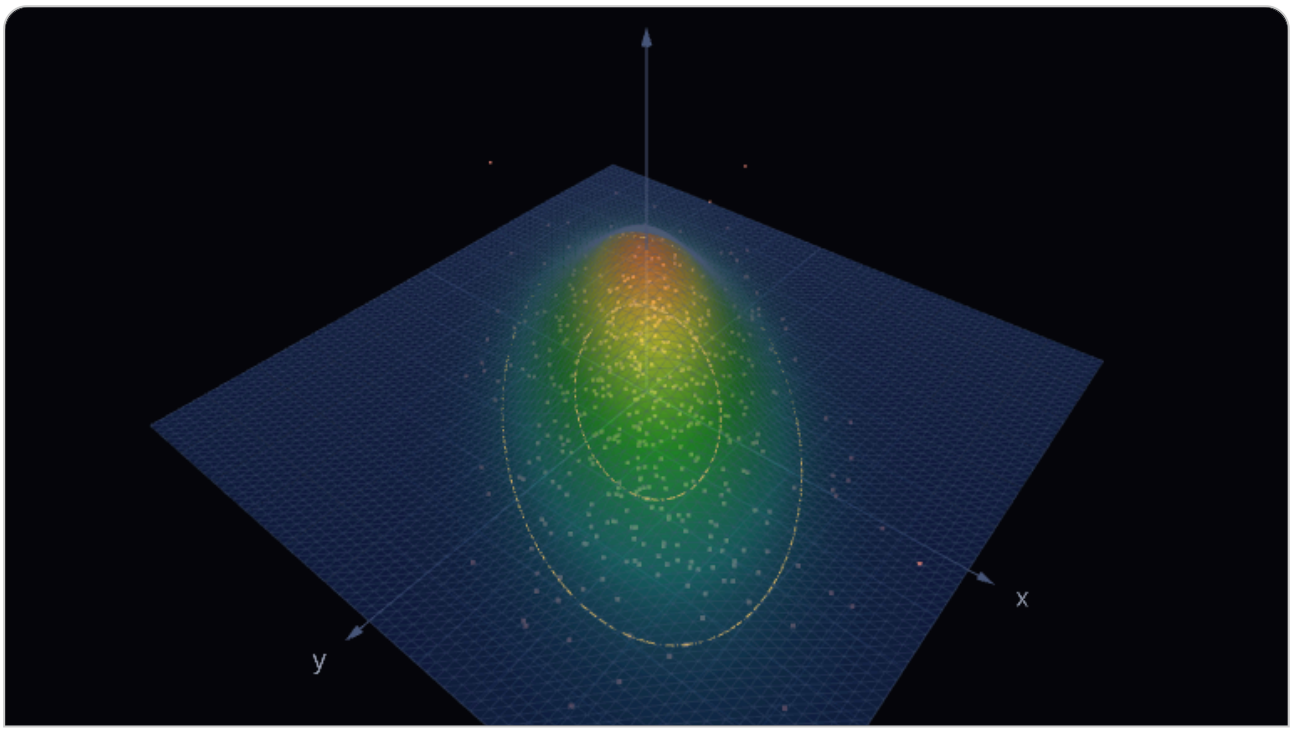
Two variables and correlation

Often two quantities are measured on the same individual: height and weight, the heights of parents and children, temperature and electricity consumption. Besides means and standard deviations, the two-dimensional normal distribution has a *correlation coefficient* ρ between −1 and 1, which measures how closely the two variables move together:

$$\rho = \frac{E\big[(X-\mu_x)(Y-\mu_y)\big]}{\sigma_x \sigma_y}$$

In 1886 Galton discovered that tall parents do indeed have tall children, but that the children are on average closer to the mean height than their parents. He called it "regression towards mediocrity" – today *regression to the mean* – and it is a pure consequence of the correlation being less than 1. The phenomenon still deceives: an athlete who has had an unusually good season will typically do worse the following year, without anything having changed.

The two-dimensional normal distribution (3D)

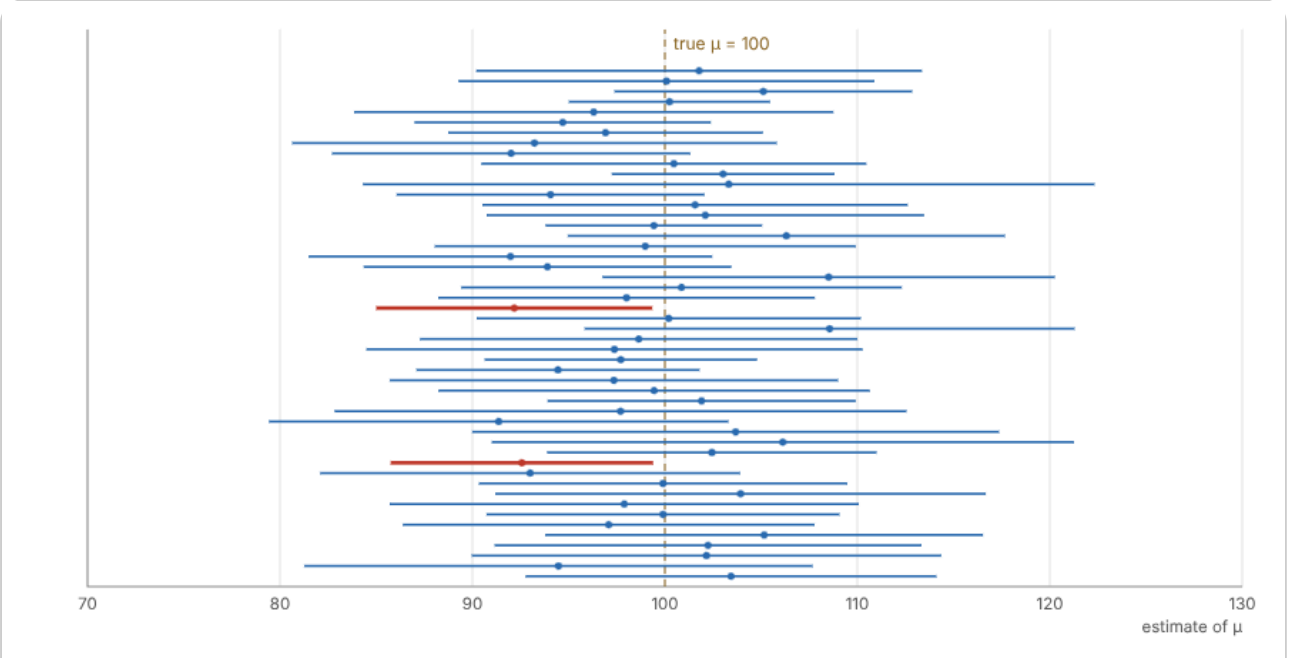


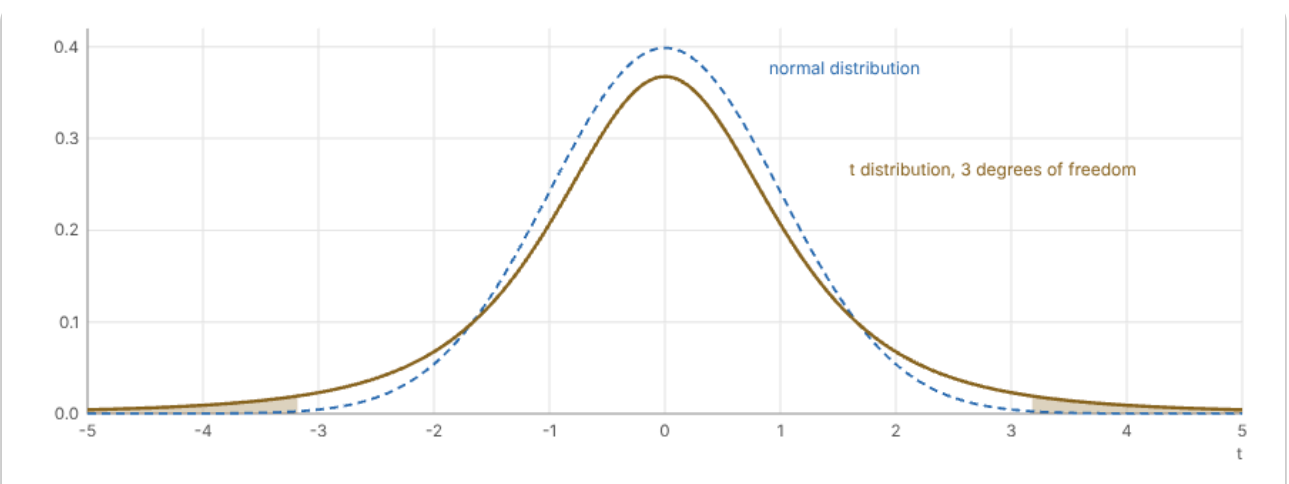
From sample to conclusion: confidence intervals and the t distribution

In practice μ is unknown; we have n measurements with average $\bar{x}$ and sample standard deviation s. A *confidence interval* is a recipe which, in a fixed proportion of cases (typically 95 %), gives an interval that contains the true value:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

Because s is itself uncertain, a slightly wider distribution than the normal must be used. This was found by the chemist William Sealy Gosset, who worked for the Guinness brewery in Dublin and had to draw conclusions from small samples of barley and hops. He published the result in 1908 under the pseudonym "Student", because Guinness did not want to reveal to its competitors that statistics was useful. Jerzy Neyman formulated the theory of confidence intervals in 1937.





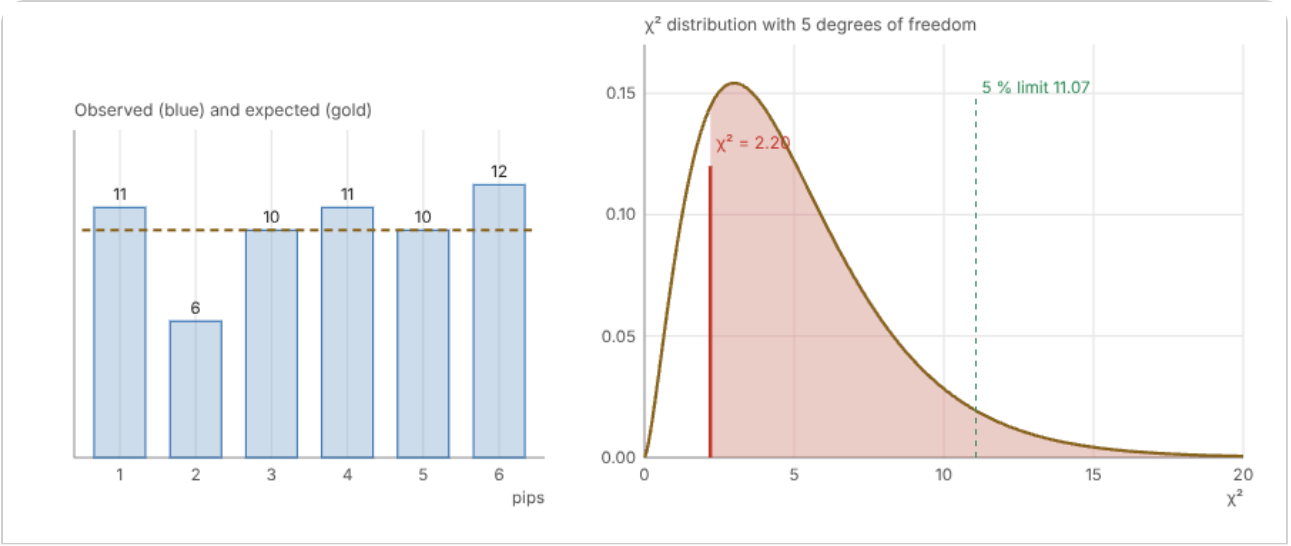
The χ² test: do the data fit the model?

In 1900 Karl Pearson introduced a way of measuring how well observed counts O_i agree with the expected E_i:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

If the model is right, χ^2 follows a known distribution, the χ² distribution with $k - 1$ degrees of freedom, where k is the number of categories. The *p-value* is the probability of getting a χ^2 at least as large if the model is right. If it is small – traditionally below 5 % – the model is rejected. A p-value is not the probability that the model is wrong, and the 5 % limit is a convention: test 20 true hypotheses and on average one will be wrongly rejected. Particle physics therefore

demands 5σ, i.e. $p < 3 \cdot 10^{-7}$, before speaking of a discovery, as when the Higgs boson was found in 2012.



Distributions in physics

Maxwell–Boltzmann

Each velocity component of a gas molecule is normally distributed with $\sigma^2 = kT/m$, and the speed follows Maxwell's distribution (1860) $f(v) \propto v^2 e^{-mv^2/2kT}$. See Statistical mechanics.

The Boltzmann factor

The probability of a state with energy E is proportional to $e^{-E/kT}$ – an exponential distribution in energy. It determines chemical reaction rates, the pressure of the atmosphere with height and the population of atomic levels.

Bose–Einstein and Fermi–Dirac

Quantum particles are indistinguishable. Bosons follow $1/(e^{(E-\mu)/kT} - 1)$ – Planck's radiation law and the laser; fermions follow $1/(e^{(E-\mu)/kT} + 1)$ – the electrons of metals and white dwarfs.

Lorentz = Cauchy

The shape of a spectral line from an atom with a finite lifetime, and the resonance curve of an oscillation, is a Cauchy distribution. Its lack of a mean is why one quotes the full width at half maximum of a line instead of a standard deviation.

Lognormal

When random contributions are multiplied instead of added, the logarithm becomes normally distributed: grain sizes in sand and dust, drop sizes in rain, incomes and share prices.

Power laws

Earthquakes follow the Gutenberg–Richter law: for each step up the Richter scale there are about ten times fewer. Heavy tails of this kind are also found in the sizes of meteorites, solar flares and the populations of cities – and they make averages unreliable, as for the Cauchy distribution.

Further reading

- David Freedman, Robert Pisani and Roger Purves: *Statistics* (4th ed., Norton 2007) – the clearest introduction without heavy formulae
- Sheldon Ross: *A First Course in Probability* (Pearson, many editions)
- Philip R. Bevington and D. Keith Robinson: *Data Reduction and Error Analysis for the Physical Sciences* (3rd ed., McGraw-Hill 2003)
- Stephen M. Stigler: *The History of Statistics* (Harvard 1986)
- David Salsburg: *The Lady Tasting Tea – How Statistics Revolutionized Science in the Twentieth Century* (Freeman 2001)
- The pages on statistical mechanics and nuclear physics

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